

Tables and fields used from the SRS and MASK MDBs

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Document generated from source code (`src/srs_reader.py`, `src/mask_converter.py`, `streamlit_app/lib/srs_inspect.py`). Lists the tables and fields actually read from each Microsoft Access database (`.mdb`) via `mdb-export` (`mdbtools`) or `ODBC` (`pypyodbc`).

Reading conventions:

- Every row is read as text and converted by `_parse_int` / `_parse_float` / `_parse_bool`. Quotes and whitespace are stripped (`.strip().strip('\"')`).
- `f_*` flags use `_parse_bool`: `Y/YES/TRUE/1` → `true`.
- Periods come in separate columns (`day/hour/min/sec`) and are summed to seconds.

1. SRS MDB (ITU/SRS orbital data)

Main NGSO filing database. Tables used: `non_geo`, `orbit`, `phase`, `sat_oper`, `mask_info`, `grp`, `mask_lnk1`.

1.1 non_geo — NGSO system/notice

Functions: `read_srs_mdb`, `list_non_geo_systems`.

Field	Parsed type	Use
<code>ntc_id</code>	<code>str</code>	Notice identifier; join key across tables
<code>sat_name</code>	<code>str</code>	Satellite/system name
<code>adm / admin</code>	<code>str</code>	Administration (country)
<code>ref_body</code>	<code>str</code>	Reference body (<code>T</code> = Earth)
<code>nbr_plane</code>	<code>int</code>	Number of orbital planes
<code>nbr_sat_td</code>	<code>int</code>	Total active satellites (density field; ≠ sats/plane) Density

density	float	(sats/km ²)
avg_dist	float	Average distance between satellites
f_x_zone	bool	Exclusion zone flag
x_zone	float	Exclusion zone angle (°)
f_constell	bool	Constellation flag
d_rcv, d_not, date_rcv, rcv_date, ntc_date, notice_date, d_sub, d_ntc, d_latest	date	Date column to sort notices most-recent-first (first present/parseable one is used)

1.2 orbit — orbital planes

Function: read_srs_mdb. Filtered by ntc_id.

Field	Parsed type	Use
ntc_id	str	System filter
orb_id	int	Orbital plane identifier
nbr_sat_pl	int	Satellites per plane (real sats/plane)
right_asc	float	RAAN — right ascension of ascending node (°)
inclin_ang	float	Inclination (°)
prd_ddd, prd_hh, prd_mm	float	Orbital period (day, hour, min) → seconds
apog, apog_exp	float, int	Apogee = apog × 10 ^{apog_exp} (km altitude)
perig, perig_exp	float, int	Perigee = perig × 10 ^{perig_exp} (km altitude)
perig_arg	float	Argument of perigee (°)
op_ht, op_ht_exp	float, int	Operational altitude = op_ht ×

		10 ^{op} _ht_exp (km)
f_stn_keep	bool	Station keeping flag
rpt_prd_dd, rpt_prd_hh, rpt_prd_mm, rpt_prd_ss	float	Ground- track repeat period → seconds
f_precess	bool	Precession flag
precession	float	Precession rate (°/day)
long_asc	float	Longitude of ascending node (°)
keep_rnge	float	Station keeping range (°)
f_sunsynch	bool	Sun- synchronous flag

1.3 phase — initial phases per satellite

Function: read_srs_mdb. Optional; absence → uniform phase per plane.

Field	Parsed type	Use
ntc_id	str	System filter
orb_id	int	Orbital plane
orb_sat_id	int	Satellite index within plane
phase_ang	float	Initial phase (°), normalized mod 360
d_ref, t_ref	int	Reference date/time; duplicate tie-break (largest wins)

1.4 sat_oper — MAX_CO_FREQ per latitude band

Function: read_sat_oper (S.1503-4 Annex D, Step 19).

Field	Parsed type	Use
ntc_id	str	System filter
		Band start latitude

lat_fr	float	(°), default -90
lat_to	float	Band end latitude (°), default +90
nbr_op_sat	int	Max simultaneous co-frequency satellites
min_dur / min_duration	float	MIN_DURATION; if ≠ 0 → NotImplementedError (§D.5.1.4.2 not implemented)

1.5 mask_info — mask metadata

Functions: read_mask_info, read_mask_assignment_all (f_mask filter).

Field	Parsed type	Use
ntc_id	str	System filter
mask_id	int	Mask identifier
freq_min	float	Minimum frequency (GHz)
freq_max	float	Maximum frequency (GHz)
f_mask	str	Type: P = PFD, E = EIRP, S = other
f_mask_type	str	Subtype: A = alpha, 0 = other

1.6 grp — emission/reception groups

Functions: read_group_for_mask, read_mask_assignment_*, srs_inspect.

Field	Parsed type	Use
ntc_id	str	System filter
grp_id	int	Group identifier
emi_rcp	str	Emission/reception (E = transmit); prioritizes mask link
beam_name	str	Beam name
freq_min	float	Minimum frequency (MHz → GHz)
freq_max	float	Maximum frequency (MHz → GHz)
elev_min	float	Minimum elevation (°)

1.7 mask_lnk1 — orbit/satellite → mask link

Functions: read_mask_assignment, read_mask_assignment_all, read_mask_assignment_per_sat, read_group_for_mask.

Field	Parsed type	Use
ntc_id	str	System filter
grp_id	int	Associated group (cross-references grp.emi_rcp for priority)
orb_id	int	Orbital plane; -1 = wildcard (applies to all)
sat_orb_id	int / blank	Specific satellite; blank/0 = all sats of the orbit
mask_id	int	Linked mask
seq_no	int	Sequence; ordering tie-break

2. MASK MDB (PFD masks)

Separate database holding the XML content of the masks. Table used: masks.

2.1 masks — PFD/EIRP masks

Functions: read_pfd_mask_xml_from_mdb, load_pfd_masks_for_ids, list_pfd_masks_from_mask_mdb.

Field	Parsed type	Use
ntc_id	str	System filter
mask_id	int	Mask identifier
f_mask	str	Type; listing keeps only P (PFD)
f_mask_type	str	Mask subtype
sat_name	str	Satellite name (label)
mask	BLOB (OLE)	Binary payload: ZIP (PK\x03\x04) containing the <pfd_mask> XML; falls back to raw

The mask payload is decoded as latin-1, the ZIP is opened in memory, and the first internal .xml is extracted and parsed by `load_pfd_mask_from_xml_content`.

Appendix — EPFD limits MDB (separate)

Not SRS nor MASK, but read through the same path
(`read_epfd_limits_from_mdb`, `suggest_services_from_limits_mdb`). Tables:
`rf_epfd_mask` (`mask_argmt`, `rr_service`, `link_direction`, `rf_band_wdth`,
`freq_min`, `freq_max`, `rf_diam`, `mask_id`, `rr_reference`) and `epfd_time`
(`mask_id`, `mask_step`, `epfd`, `time_percent`, `rr_reference`).